Procedure

1. Open Android Studio IDE → File → New → New Project → specify the application name "SD Card" and company domain "com.mad.sdcard" → click "next" → choose Minimum SDK "API 17:Android 4.2(Jelly Bean)" → click "Next" → choose "Blank Activity" → click "next" → specify the Activity Name "MainActivity" → click "Finish".
2. Open MainActivity.java under app/java/ sdcard.mad.com.sdcard and type the following codes:

MainActivity.java

```
package sdcard.mad.com.sdcard;

import android.app.Activity;
import android.app.Notification;
import android.app.NotificationManager;
import android.app.PendingIntent;
import android.content.Intent;
import android.os.Bundle;
import android.os.Environment;
import android.os.Handler;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import java.io.BufferedReader;
import java.io.File;
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.InputStreamReader;
import android.support.v4.app.NotificationCompat;

public class MainActivity extends Activity {

    EditText editText;
    TextView textView;

    Button writeBtn, readBtn, notifyBtn, timerBtn, clearBtn;

    String filename = "sample.txt";

    @Override
    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editText = (EditText) findViewById(R.id.editText);
```

```
textView = (TextView) findViewById(R.id.textView);

writeBtn = (Button) findViewById(R.id.btnWrite);
readBtn = (Button) findViewById(R.id.btnRead);
notifyBtn = (Button) findViewById(R.id.btnNotify);
timerBtn = (Button) findViewById(R.id.btnTimer);
clearBtn = (Button) findViewById(R.id.btnClear);   // CLEAR BUTTON INITIALIZED

// WRITE FILE TO SD CARD
writeBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        try {

            File path = Environment.getExternalStorageDirectory();
            File file = new File(path, filename);

            FileOutputStream fos = new FileOutputStream(file);
            fos.write(editText.getText().toString().getBytes());
            fos.close();

            Toast.makeText(MainActivity.this,
                    "File Written to SD Card",
                    Toast.LENGTH_LONG).show();

        } catch (Exception e) {
            e.printStackTrace();
        }

    }
});

// READ FILE FROM SD CARD
readBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        try {

            File path = Environment.getExternalStorageDirectory();
            File file = new File(path, filename);

            FileInputStream fis = new FileInputStream(file);

            BufferedReader br = new BufferedReader(
                    new InputStreamReader(fis));

            String line;
            String data = "";
```

```
            while ((line = br.readLine()) != null) {
                data = data + line;
            }

            br.close();

            textView.setText(data);

            Toast.makeText(MainActivity.this,
                    "File Read Successful",
                    Toast.LENGTH_LONG).show();

        } catch (Exception e) {
            e.printStackTrace();
        }

    }
});

// CLEAR TEXT BUTTON
clearBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        editText.setText("");
        textView.setText("");

        Toast.makeText(MainActivity.this,
                "Text Cleared",
                Toast.LENGTH_SHORT).show();
    }
});

// PERSISTENT NOTIFICATION
notifyBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        NotificationManager nm =
                (NotificationManager) getSystemService(NOTIFICATION_SERVICE);

        Intent intent = new Intent(MainActivity.this, MainActivity.class);

        PendingIntent pi = PendingIntent.getActivity(
                MainActivity.this, 0, intent, 0);

        Notification notification = new NotificationCompat.Builder(MainActivity.this)
                .setSmallIcon(R.mipmap.ic_launcher)
                .setContentTitle("Application Running")
```

```
                .setContentText("This notification cannot be cleared")
                .setTicker("Persistent Notification")
                .setContentIntent(pi)
                .setOngoing(true)
                .build();

            nm.notify(1, notification);

        }
    });

    // TIME BASED NOTIFICATION (5 seconds delay)
    timerBtn.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {

            Handler handler = new Handler();

            handler.postDelayed(new Runnable() {
                @Override
                public void run() {

                    NotificationManager nm =
                        (NotificationManager) getSystemService(NOTIFICATION_SERVICE);

                    Intent intent = new Intent(MainActivity.this, MainActivity.class);

                    PendingIntent pi = PendingIntent.getActivity(
                        MainActivity.this, 0, intent, 0);

                    Notification notification = new
NotificationCompat.Builder(MainActivity.this)
                        .setSmallIcon(R.mipmap.ic_launcher)
                        .setContentTitle("Reminder")
                        .setContentText("Notification after 5 seconds")
                        .setTicker("Timer Notification")
                        .setContentIntent(pi)
                        .build();

                    nm.notify(2, notification);

                }
            }, 5000);

        }
    });

  }

}
```

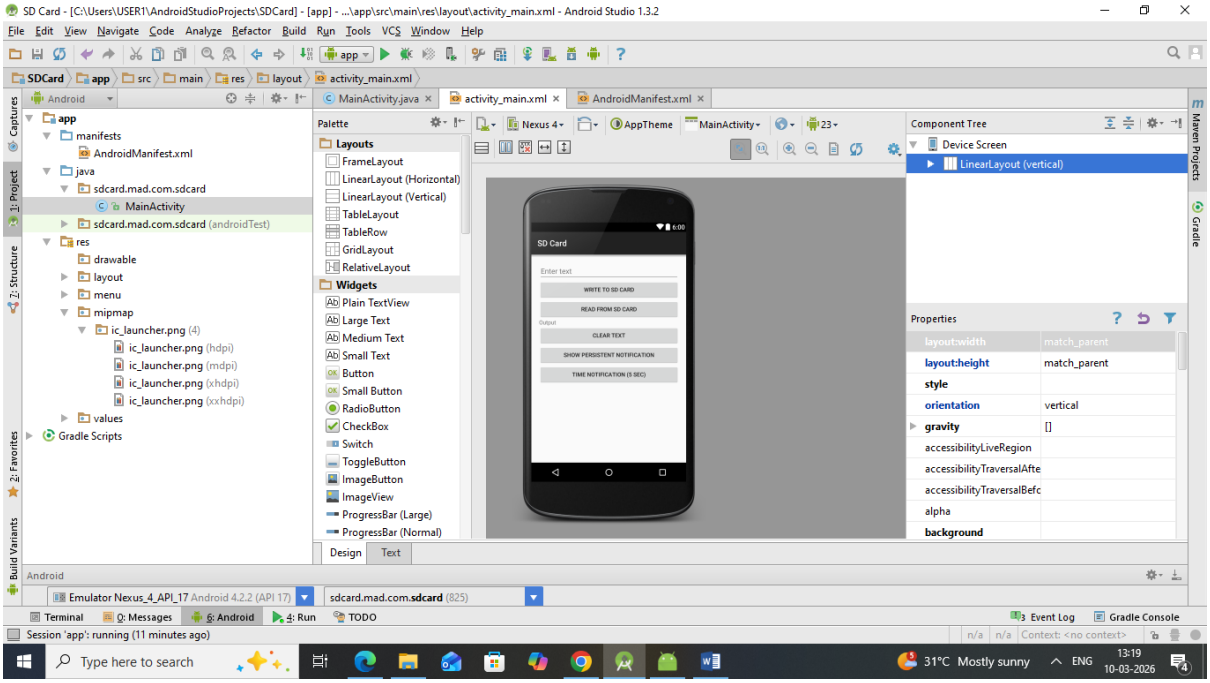
3. Open activity_main.xml under app/res/layout and type the following codes:

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/editText"
        android:hint="Enter text"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnWrite"
        android:text="Write to SD Card"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnRead"
        android:text="Read from SD Card"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/textView"
        android:text="Output"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/btnClear"
        android:text="Clear Text"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/btnNotify"
        android:text="Show Persistent Notification"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnTimer"
        android:text="Time Notification (5 sec)"
        android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"/>
</LinearLayout>
```

4. The design of the application will be as follows:



5. Open AndroidManifest.xml under app/manifests folder and do the following changes:

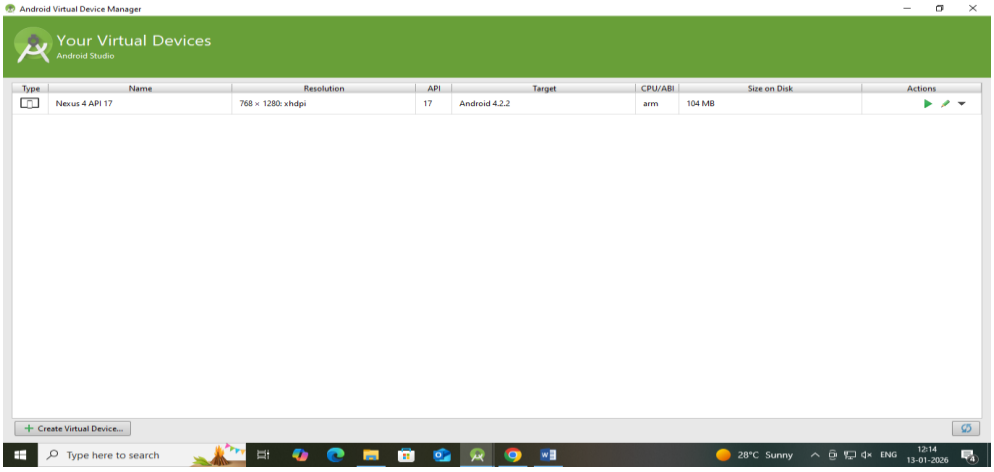
AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
   package="sdcard.mad.com.sdcard" >

   <uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
   <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE"/>

   <application
      android:allowBackup="true"
      android:icon="@mipmap/ic_launcher"
      android:label="@string/app_name"
      android:theme="@style/AppTheme" >
      <activity
         android:name=".MainActivity"
         android:label="@string/app_name" >
         <intent-filter>
            <action android:name="android.intent.action.MAIN" />

            <category android:name="android.intent.category.LAUNCHER" />
         </intent-filter>
```

```
    </activity>
  </application>

</manifest>
```

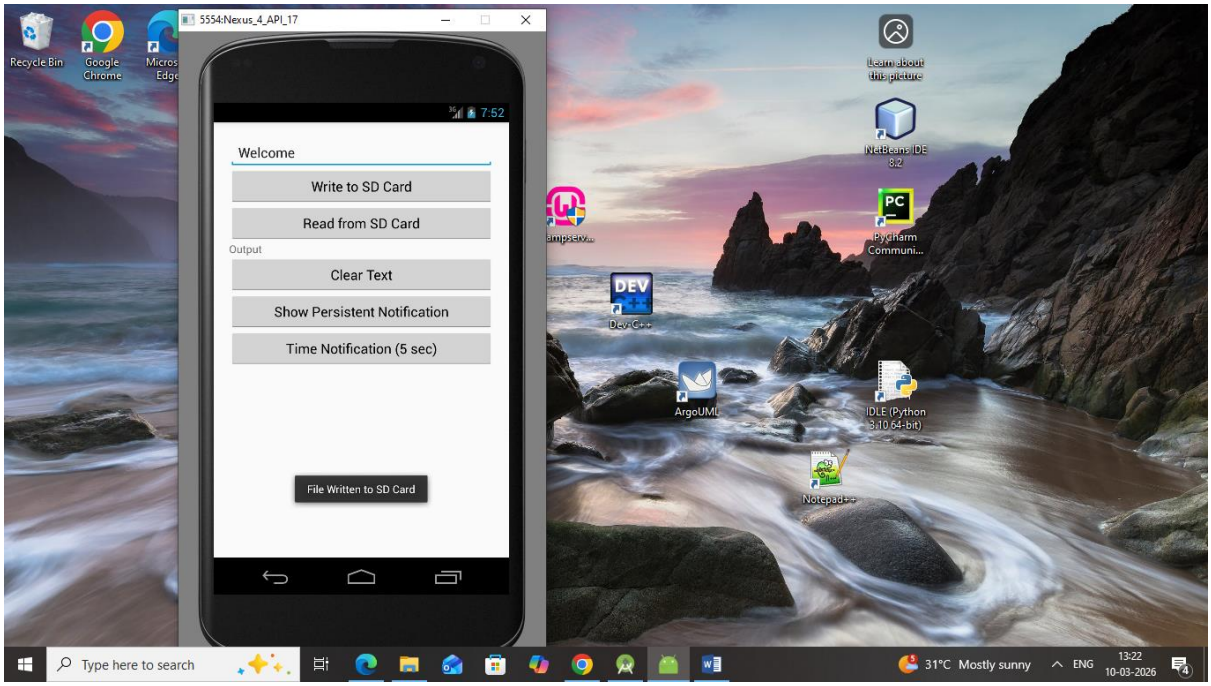
6. Go to Tools → android → AVD Manager → click "+ create a virtual device" → select "phone" from category → select "Nexus 4" from the list → click "next" → select Release name: Jelly Bean, API Level: 17, ABI: armeabi-v7a, Target: Android 4.2.2 from the list → click "next" → Choose orientation "portrait" → click "finish".

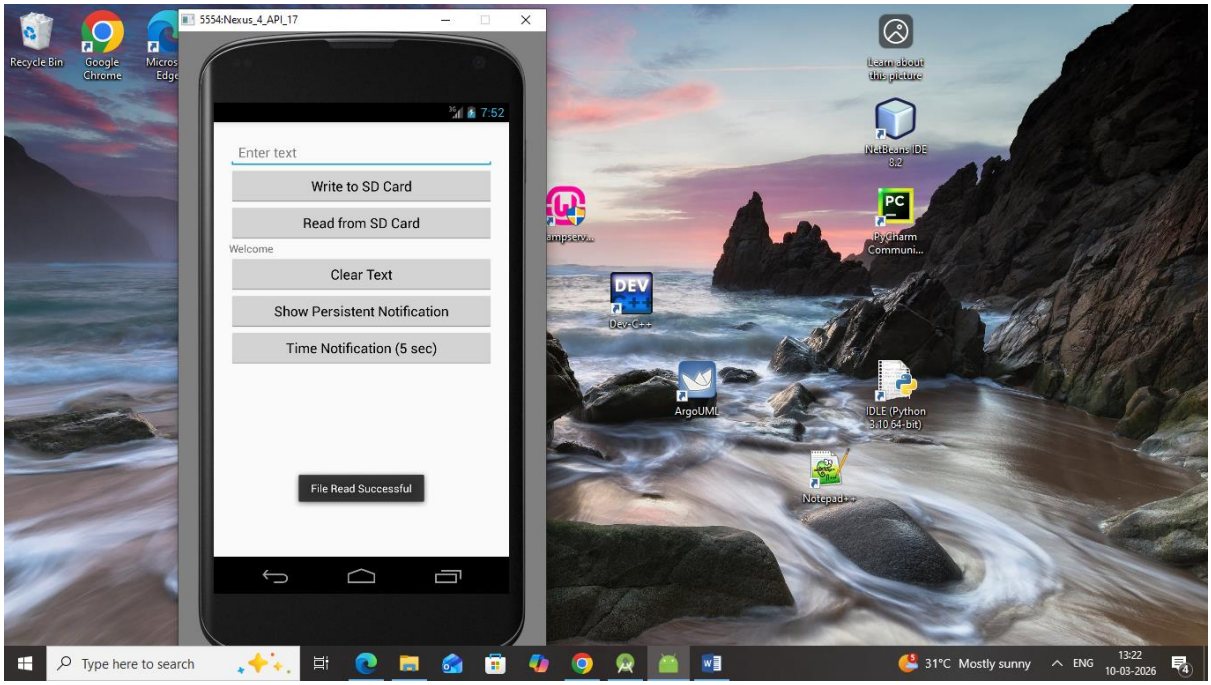
The following window will appear after configuring AVD:

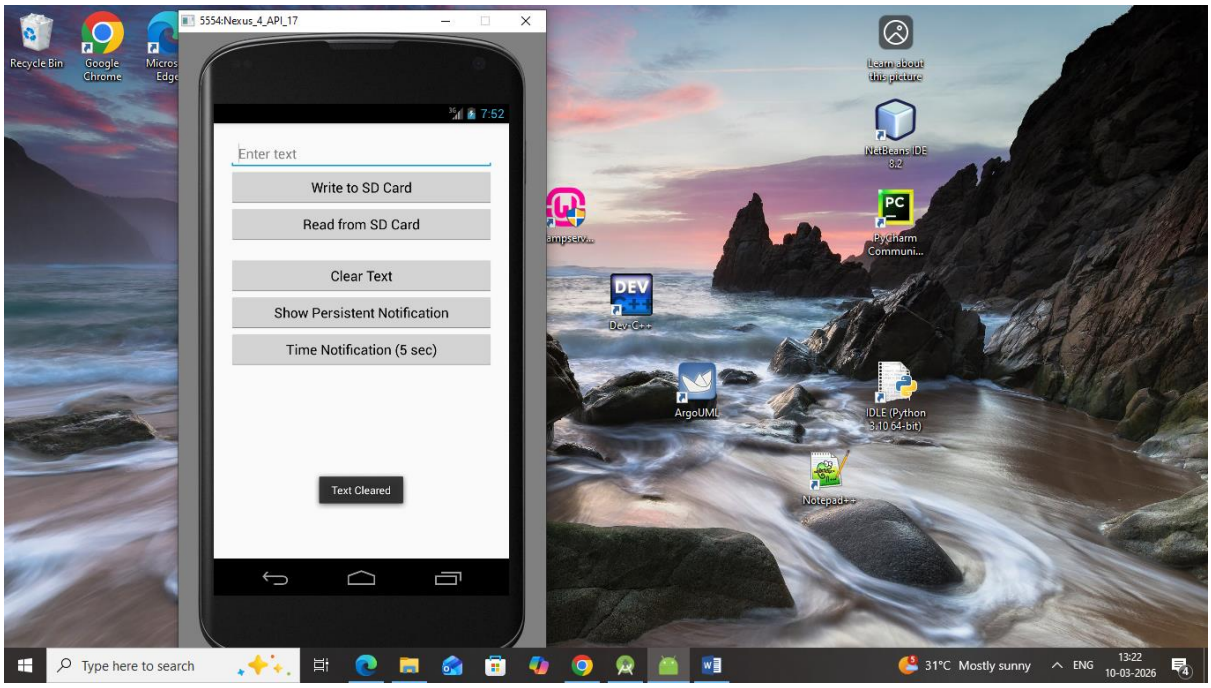


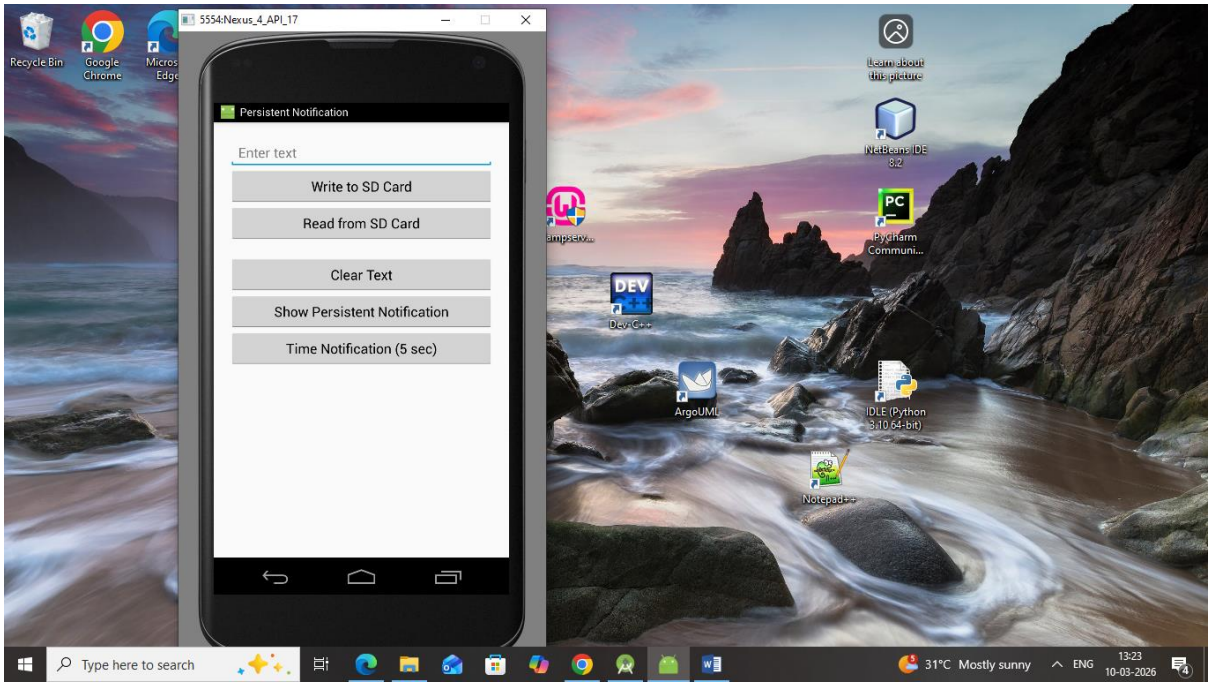
7. Click "Run app" button in the Android Studio → choose android virtual device → click "ok".

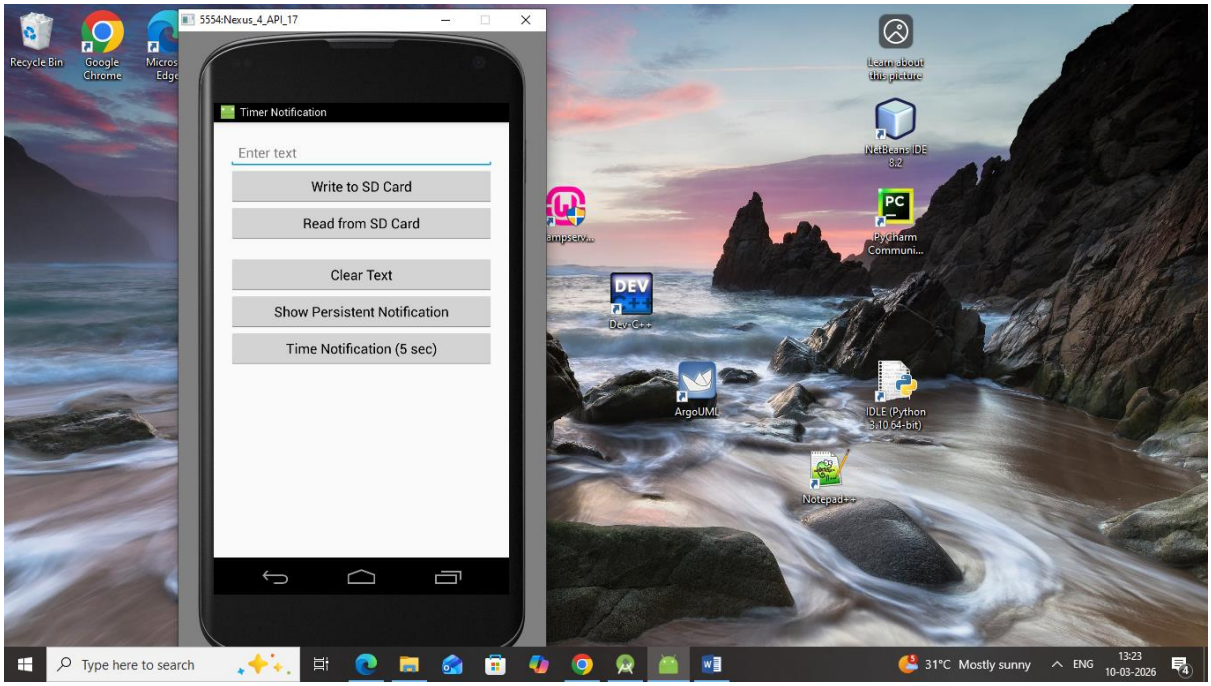
Output

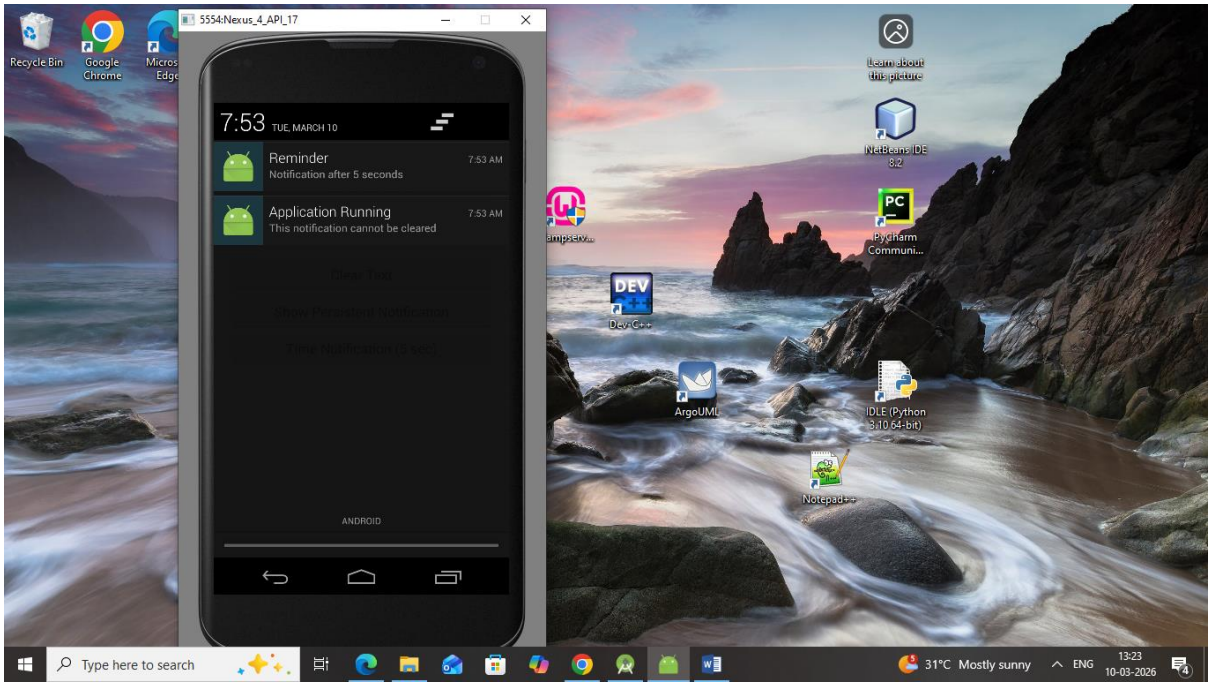



5554:Nexus_4_API_17
7:52
Enter text
Write to SD Card
Read from SD Card
Welcome
Clear Text
Show Persistent Notification
Time Notification (5 sec)
File Read Successful


5554:Nexus_4_API_17
7:52
Enter text
Write to SD Card
Read from SD Card
Clear Text
Show Persistent Notification
Time Notification (5 sec)
Text Cleared


5554:Nexus_4_API_17
Persistent Notification
Enter text
Write to SD Card
Read from SD Card
Clear Text
Show Persistent Notification
Time Notification (5 sec)


5554:Nexus_4_API_17
Timer Notification
Enter text
Write to SD Card
Read from SD Card
Clear Text
Show Persistent Notification
Time Notification (5 sec)


5554:Nexus_4_API_17
7:53 TUE, MARCH 10
Reminder
Notification after 5 seconds
7:53 AM
Application Running
This notification cannot be cleared
7:53 AM
ANDROID
Recycle Bin
Google Chrome
Learn about this picture
NetBeans IDE 8.2
PyCharm Communi...
Dev-C++
ArgoUML
IDLE (Python 3.10 64-bit)
Notepad++
Type here to search
31°C Mostly sunny
ENG
13-23
10-03-2026